25901184 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BCA (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Skill Enhancement Course (SEC) - MG4SECBCA200 - OBJECT ORIENTED PROGRAMMING USING JAVA

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **3** marks

1. What is Java Virtual Machine (JVM)? [K] / [CO1]
2. How does a switch statement simplify multiple decision-making in a program? [U] / [CO1]
3. How will you check whether a Vector is empty or not? [An] / [CO2]
4. Explain abstract methods in Java [An] / [CO2]
5. Define a package [U] / [CO3]

[3x5 = 15]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **5** marks

6. Compare static variables and instance variables. [U] / [CO1]

7. Describe method overloading in Java and demonstrate its implementation with a suitable example. [U] / [CO1]

8. Create and display elements of an n X n matrix [An] / [CO2]

9. Explain how to prevent overriding a variable in Java. Give an example. [U] / [CO2]

10. Explain the steps involved in creating a thread by extending the Thread class with a simple example. [An] / [CO3]

[5x4 = 20]

Part C

Essay Questions

Answer any **1** questions

Each question carries **15** marks

11. With the help of suitable examples explain Wrapper methods. [A] / [CO2]

12. Create a class Student with attributes roll no, name, age and course. If age of student is not in between 15 and 21 then generate user-defined exception “AgeNotWithinRangeException”. [A] / [CO3]

[15x1 = 15]